

Vojtěch Vlček

AI engineer



Summary

Python and AI engineer based in Prague, building LLM-driven workflows, provider integrations, and practical software for clients and research teams. Work with LangGraph, Pydantic AI, LangSmith, OpenAI and Gemini APIs, FastAPI, PostgreSQL/SQLite, Docker, and Linux. Focus on maintainable Python, structured outputs, validation, testing, observability, and resumable execution across AI automation, media generation, and GPU training systems.

Professional Experience

Artificial Intelligence Engineer, RESULMATIC

- Develop a next-generation AI agent that helps people find cars matching their needs and preferences.
- Build with Python, LangGraph, Pydantic AI and LangSmith. Work with Kubernetes, PostgreSQL and Microsoft Azure, including Azure Foundry and Azure developer services.

07/2026 – Present
Prague, Czechia | Hybrid

Applied AI & Deep Learning Engineer, 3BrainAI Nexus

- Apply AI and deep learning to construction-risk evaluation.

04/2026 – Present
Remote | Contract

Freelance AI Engineer, Upwork Clients

- Built a Python/FastAPI application that turns marketing briefs into ad images and short videos, integrating Gemini, Google image generation, Veo, and Kling.
- Implemented prompt pipelines, background generation with status polling, and automatic logo placement on images and videos.
- Delivered AI-media projects for Upwork and independent clients, with 5.0-rated project outcomes.

2025 – 07/2026
Remote

Primary Researcher, Sign Language Synthesis, University of West Bohemia

- Implemented a PyTorch/Diffusers pipeline for pose-conditioned sign-language generation, including data preparation, GPU training, checkpointing, and evaluation.
- Added OpenCV/HDF5 data loading, quality filtering, W&B logging, and Docker/Singularity environments.
- University-affiliated mentored project, outside formal employment.

09/2024 – 09/2025
Pilsen, Czechia

Contract Python Developer, Czech Academy of Sciences

- Built a Python CLI to convert experimental XML exports into legacy trajectory files, preserving event timing through event-to-frame alignment.
- Added input validation, deterministic file naming, malformed-file diagnostics, and fixed-width export.

05/2025 – 06/2025
Prague, Czechia

Contract Researcher, University of West Bohemia

- Built Python/OpenCV tools to classify camera movement and compare recordings against reference videos.
- Implemented optical-flow analysis, SSIM damaged-frame detection, and parallel video processing for recording-quality checks.

09/2024 – 01/2025
Pilsen, Czechia

Contract Python Developer, Czech Academy of Sciences

- Built, tested, and debugged a Python memory-test application for a Czech Academy of Sciences research team.

01/2022 – 06/2022
Prague, Czechia

</> Technical Skills

Programming languages

- Python, Java, C#

Development & infrastructure

- Linux, Docker, Git, pytest, CUDA, Kubernetes, W&B, Langfuse

AI & machine learning

AI agents, LLM/RAG pipelines, PyTorch, Diffusers, OpenCV, model training and evaluation, generative media

Backend & data

- FastAPI, PostgreSQL, SQLite, pgvector, data conversion/ETL, Flutter, FFmpeg/MoviePy

📁 Personal Projects

CleanCodeRefactor [🔗](#)

- Built a Python CLI for Codex-assisted refactoring in isolated copied Git workspaces; the original repository remains unchanged until changes are reviewed and explicitly applied with ccr apply --yes.
- Extracted Python refactoring units with Tree-sitter and an AST fallback; added maintenance-value ranking, include/exclude filters, and token-budgeted clustering to select work and fit relevant context into model prompts.
- Orchestrated Pydantic-validated Codex CLI calls for reference retrieval, test auditing and writing, refactoring, and change review. Managed prompts and traces through Langfuse.
- Built a verification loop with characterization baselines, formatting, compile and pytest checks, retry feedback, rollback, and per-unit Git commits. Persisted run state and JSONL logs for resumption, with dashboards and focused diffs for review.

AutoChannel [🔗](#)

- Built a Python pipeline that turns SQLite channel and topic records into scripts, narration, generated images or video clips, subtitles, thumbnails, publishing metadata, final MP4 renders, and optional YouTube uploads.
- Designed pluggable LLM, speech, image, and video handlers integrating OpenAI, Gemini, OpenRouter, Cartesia, Veo, Kling, and other providers; supported structured LLM output and Langfuse-managed prompts.
- Defined schema-validated editorial rhythm maps to guide pacing and visual assembly. Aligned generated narration with Whisper transcriptions to produce word-timed subtitles.
- Added SQLite stage checkpoints, content-based invalidation, and resumable generation. Used MoviePy and FFmpeg for assembly, with Docker setup and pytest coverage for checkpoints, prompts, handlers, and rendering.
- Built a channel-research workflow that combines video metadata, sampled frames, and multimodal LLM analysis into structured channel profiles and automation assessments.

MuAlphaZeroLibrary [🔗](#)

- Built a reusable PyTorch reinforcement-learning library with abstract game, network, memory, arena, and Monte Carlo tree search interfaces for AlphaZero- and MuZero-style training.
- Implemented self-play, replay buffers, parallel workers, checkpoint/resume flows, and model pitting; added W&B experiment logging and Optuna hyperparameter search for long-running training.

JavaTorch [🔗](#)

- Built a Java 23/Maven deep-learning framework on ND4J with PyTorch-style tensors, manually implemented autograd, neural-network modules, losses, and an Adam optimizer. Implemented convolution with im2col/col2img gradient reversal, computation-graph cleanup and visualization, and demo CNN training flows.

🎓 Education

Computer Science, University of West Bohemia

- Completed the first year of a bachelor's program before moving into software engineering and client work. Degree not completed.

08/2024 – 08/2025

Pilsen, Czechia

High School Diploma, Gymnázium Da Vinci

2020 – 2024

Czechia

Courses

Neuromatch Academy in Deep Learning , *Neuromatch Academy*

07/2023 – 07/2023

- Completed an intensive three-week program in deep learning with hands-on neural-network implementation.

Machine Learning Specialization , *Coursera*

10/2022 – 11/2022

- Completed coursework in supervised learning, neural networks, and model evaluation.

Languages

English

C2

Czech

Native

Interests

Programming, AI, 3D modeling, storytelling, nature and animals